EGE UNIVERSITY

FACULTY OF ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

PROGRAMMING LANGUAGES

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PROJECT-1 REPORT

The Plus++ Language Lexical Analyzer

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2) INPUTS AND OUTPUTS

INPUTS

```
number size
number sum
size := 5
repeat size times
{
    write size and newLine
    sum += size
}
write newLine and "Sum:" and sum
```

OUTPUTS

```
Keyword(number)
Identifier(size)
EndOfLine
Keyword(number)
Identifier(sum)
EndOfLine
Identifier(size)
Operator(:=)
IntConstant(5)
EndOfLine
Keyword(repeat)
Identifier(size)
Keyword(times)
OpenBlock
Keyword(write)
Identifier(size)
Keyword(and)
Keyword(newline)
EndOfLine
Identifier(sum)
Operator(+=)
Identifier(size)
EndOfLine
CloseBlock
Keyword(write)
Keyword(newline)
Keyword(and)
StringConstant("Sum:")
Keyword(and)
Identifier(sum)
EndOfLine
```

3) ERROR CHECKING

Token too long error

```
Keyword(number)
Error: Unrecognized token 'verylongidentifiernamethatisdefinitelylongerthanallowedlimit'.
```

Unrecognized token error

```
Keyword(number)
Error: Unrecognized token 'x$'.
```

Unclosed string constant error

```
Keyword(write)
Error: Unclosed string constant.
```

Invalid operator or usage error

```
Identifier(x)
Operator(:=)
IntConstant(5)
EndOfLine
Identifier(y)
Error: Unrecognized token '+' without '='.
```